Project Title	**Education Regulation Impact Analyzer (ERIA)** ***Simplifying Education Policies and Circulars for Every Institution***
Skills take away From This Project	• **Web scraping / Document ingestion (PDF, HTML bills from Sansad portals)** • **NLP-based text preprocessing** • **Topic extraction & legislative context mapping** • **Chronology & dependency graph generation** • **Domain/industry impact analysis** • **Short-term / Medium-term / Long-term impact assessment** • **LLM-powered summarization** • **Streamlit/Gradio based dashboard** • **Langflow workflow design**
Domain	**EdTech / Education Policy Analytics**

Problem Statement:

Education policies, university regulations, accreditation guidelines, scholarship notifications, and circulars issued by bodies such as UGC, AICTE, NAAC, NIRF, and universities are usually written in complex academic, legal, and administrative language. Because of this, students, faculty, and institutions often struggle to understand the real meaning and impact of these regulations. People often do not know:

- **What the circular or regulation actually proposes or changes**
- **Why the regulation was introduced and what problem it aims to solve**
- **How the circular relates to earlier policies, amendments, or academic frameworks**
- **Which stakeholders, institutions, departments, or learners will be affected positively or negatively**
- **What the short-term, medium-term, and long-term consequences of the regulation may be**

As a result, institutions may miss important compliance updates, students may remain unaware of opportunities or policy changes, and administrators may find it difficult to plan timely actions.

This project aims to build an AI-powered platform that allows users to upload any education regulation document (PDF, HTML, or official link). The system will automatically analyze the document and convert it into a simple, easy-to-understand, stakeholder-friendly summary. It will explain the regulation's purpose, historical background, affected stakeholders, institutional implications, opportunities, and risks across different timeframes. The platform will help people gain clear insights into education regulations without needing policy or compliance expertise.

Business Use Cases

1. Simplified Regulation Summaries

Convert long education policy documents into simple, student/faculty/admin-friendly summaries.

2. Chronology Analyzer

Identify:

- Historical context
- Previous related circulars
- Amendments
- Policy evolution

3. Stakeholder & Institutional Impact Mapping

Automatically detect which groups the regulation affects, e.g.:

- Students
- Faculty
- Colleges
- Universities
- Academic administrators
- Accreditation teams
- Scholarship applicants

4. Impact Forecast

Provide structured analysis:

- Short-term impact
- Medium-term impact
- Long-term impact
- Risks & unintended consequences

5. Awareness & Academic Support

Students, faculty members, academic planners, administrators, and researchers can instantly understand the significance of new education circulars.

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6. Governance & Compliance

Institutions can understand whether upcoming regulations affect their:

- Academic operations
- Compliance requirements
- Documentation workload
- Accreditation readiness
- Policy opportunities

Approach

1. Dataset Preparation

Input Sources

- PDF circular files uploaded by the user
  Collect education regulation documents from trusted official sources such as:
- **UGC (University Grants Commission)** — regulations, circulars, scholarship notices, university-related notifications. (Sample links attached)

  https://www.ugc.gov.in/Notices?utm_source=chatgpt.com

  https://www.ugc.gov.in/regulations?utm_source=chatgpt.com

  https://www.ugc.gov.in/Circulars?utm_source=chatgpt.com

  https://www.ugc.gov.in/Guideline?utm_source=chatgpt.com

You can also get inputs from following references(optional)

- **AICTE (All India Council for Technical Education)**
- **NAAC (National Assessment and Accreditation Council)**
- **NIRF (National Institutional Ranking Framework)**
- **Ministry of Education, Government of India**

2. Text Preprocessing and Structuring

- Process and organise as per the need

3.Model Development (GEMINI API / Groq / Any api that supports the process)

Components

- **Regulation Topic Classifier**

Identify document category (Accreditation, Scholarship, Curriculum, Faculty Policy, Examination, Admissions, etc.)

- **Chronology Builder**

Detects predecessor circulars, amendments, accreditation revisions, policy background.

- **Impact Analyzer**

Extract implications for:

  - students
  - faculty
  - institutions
  - administrators
  - accreditation / compliance teams

- **Sentiment & Risk Detection**

Identify:

  - Controversial areas
  - Potential implementation issues
  - Institutional readiness challenges

4. LLM-Powered Summarization & Recommendation Generation

Outputs

1. Easy-to-read 10-20 line summary

"What does the regulation mean for a normal student, faculty member, or institution?"

2. Stakeholder Impact Report

- Which stakeholders benefit
- Which stakeholders face constraints
- New academic or institutional opportunities

3. Chronology & Policy Mapping (optional)

Visual (optional) flow of:

- Previous circulars
- Related amendments
- Committees / frameworks
- Historical evolution

4. Impact Assessment (Basic Inference)

- Short term (0–1 year)
- Medium term (1–5 years)
- Long term (>5 years)

5. Positives and Negatives

Simple bullet-point analysis.

5. Deployment (Streamlit / Gradio Dashboard)

Dashboard Features

- Upload regulation PDF / Paste URL
- AI Summary Panel
- Timeline view of policy history
- Stakeholder Impact Data or Visual
- Downloadable Summary in PDF form (optional)

Expected Results

1. Simplified regulation understanding for users

Clear explanation of intent, purpose, and impact.

2. Stakeholder-wise insights

Useful for:

- Students
- Faculty
- Institutions
- Policy researchers

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3. Risk & Readiness Detector

Identifies clauses that may:

- Increase institutional burden
- Affect academic flexibility
- Cause compliance or accreditation issues

4. Awareness & transparency

Empowers education stakeholders with reliable information.

Impact

- Bridges the gap between regulators and education stakeholders
- Enhances policy awareness in institutions
- Supports faculty, students, and administrators

- Helps institutions prepare for regulatory changes
- Improves academic planning and informed decision-making

Project Evaluation Metrics

1. Accuracy of regulation topic classification
2. Chronology correctness (optional)
3. Relevance of stakeholder impact analysis
4. Quality of LLM summarization
5. User satisfaction & readability
6. Processing time for documents

Technical Tags

- NLP
- Transformers
- Education NLP
- Langflow
- Gradio
- Knowledge Graphs
- Summarization
- Text Classification
- Regulation Analytics

Project Deliverables

1. Python Codebase (GitHub).
2. Streamlit or Gradio-based dashboard.
3. Deployed demo link (Hugging Face Spaces / Replit).
4. Documentation & README with usage instructions.
5. Video walkthrough or demo.
6. Optional: Project presentation deck.

Project Guidelines

1. **Version Control**:
    - Use Git for tracking changes.
    - Include detailed commit messages.
2. **Coding Standards**:
    - Follow PEP 8 guidelines for Python.
    - Modularize code into functions or classes.
3. **Best Practices**:
    - Test the model on unseen data before deployment.
    - Monitor class imbalance during training to avoid bias.

Timeline:

2 week

References:

Hugging face course - Link

Streamlit documentation in hugging face - Link

Pre-trained models - Link

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.
Note: Book the slot at least before 12:00 Pm on the same day

Timing: Monday-Saturday (4:00PM to 5:00PM)

Booking link :https://forms.gle/XC553oSbMJ2Gcfug9

For DE/BADM project/class topic doubt slot clarification session:

Booking link : https://forms.gle/NtkQ4UV9cBV7Ac3C8

Session timing:

For DE: 04:00 pm to 5:00 pm every saturday
For BADM 05:00 to 07:00 pm every saturday

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.
Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing:

For BADM and DE
Monday-Saturday (11:30AM to 1:00PM)

For DS and AIML
Monday-Saturday (05:30PM to 07:00PM)

Booking link : https://forms.gle/1m2Gsro41fLtZurRA

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